1. Problem Statement

Navigating intellectual property and regulatory requirements for traditional medicine, particularly Ayurveda, Siddha and Unani (ASU), is challenging because multiple national laws and international treaties may apply simultaneously. Practitioners, researchers, micro, small and medium enterprises (MSMEs) and startups often struggle to identify the applicable legal provisions and understand how different regulatory frameworks affect their formulations. Questions involving patentability and traditional knowledge alongside biodiversity-related obligations and product classification can require careful examination of several authoritative documents. General-purpose AI systems may produce inaccurate legal explanations or unsupported citations which can mislead users during regulatory research. Furthermore, lengthy statutory documents and complex legal terminology make it difficult for non-specialists to locate and interpret relevant information efficiently.

IP-SAKTI Sahayak addresses these challenges through a jurisdiction-aware and formulation-specific Retrieval-Augmented Generation (RAG) system. It uses a curated regulatory corpus and semantic retrieval to identify relevant legal evidence before generating structured guidance. By validating source references and physical PDF page mappings, the system aims to make regulatory information more accessible and traceable. It also identifies insufficient evidence rather than presenting unsupported information as established legal guidance.

2. Aim of the Mini Project

The aim of IP-SAKTI Sahayak is to develop and evaluate a source-grounded Retrieval-Augmented Generation (RAG) system that:

- **Corpus Curation and Preprocessing:** Organizes authoritative regulatory documents into structured records while preserving original legal text and verifying source metadata and physical PDF page references.
- **Semantic Retrieval and Response Generation:** Uses MiniLM embeddings and FAISS similarity search to identify relevant legal passages and provide contextual evidence to the configured Gemini model for generating structured regulatory responses.
- **Evidence Validation and System Evaluation:** Applies deterministic checks to source identifiers and exact quotations alongside page mappings and response structure. The system is evaluated through automated tests and retrieval benchmarks to assess evidence traceability and its handling of incomplete or unsupported queries.

Together, these objectives establish a structured and auditable data science foundation for intellectual property and Ayurveda-related regulatory guidance. The mini-project focuses on the core RAG pipeline and its evidence-validation mechanisms before completing the integration of the existing frontend and backend workflows.

3. Front-End / User Interface

IP-SAKTI Sahayak provides a web-based interface for accessing intellectual property guidance and Ayurveda-related regulatory information. The interface combines user authentication and guided workflows with a technology architecture designed for source-based legal information retrieval.

3.1 Home Page and Navigation

The homepage introduces IP-SAKTI Sahayak and its purpose of providing verified intellectual property guidance. The navigation bar provides access to Guidance, Formulation Classifier and Resource Library alongside Login and Register options.

3.2 User Authentication

The login and registration interface allows users to access their workspace. Authentication supports a personalized application experience for users accessing the available features.

3.3 Regulatory Guidance

Users select India or International as the jurisdiction and choose a regulatory domain such as Patents, Biodiversity or ASU Regulatory. The Quick Inquiries dropdown provides predefined questions while the input field allows users to submit queries in natural language. Reset and submission controls support successive inquiries in a chat-style interface.

3.4 Formulation Classifier

The classifier uses a four-step questionnaire covering textual origin and intended use alongside processing method and excipients or delivery systems. A progress indicator and Back and Next Step controls guide users through the questions before they submit their selections for formulation assessment.

3.5 System Architecture and Technology Stack:

The workflow connects user queries with trusted legal sources through Python and FAISS-based semantic retrieval. The configured Gemini API generates structured responses while validation mechanisms check citations and source references. The technology stack includes React and JavaScript alongside Tailwind CSS and Node.js with Express and MongoDB. Python and FAISS support the retrieval pipeline while Gemini API provides the current generation capability.

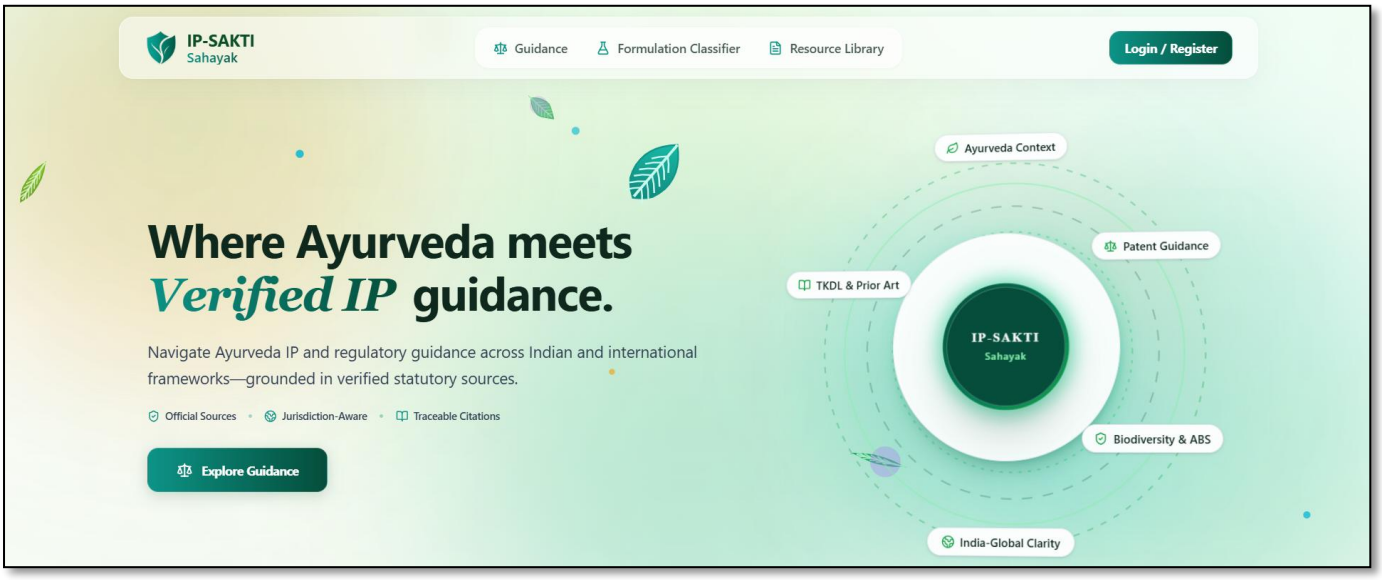


Figure 3.1: IP-SAKTI Sahayak Home Page

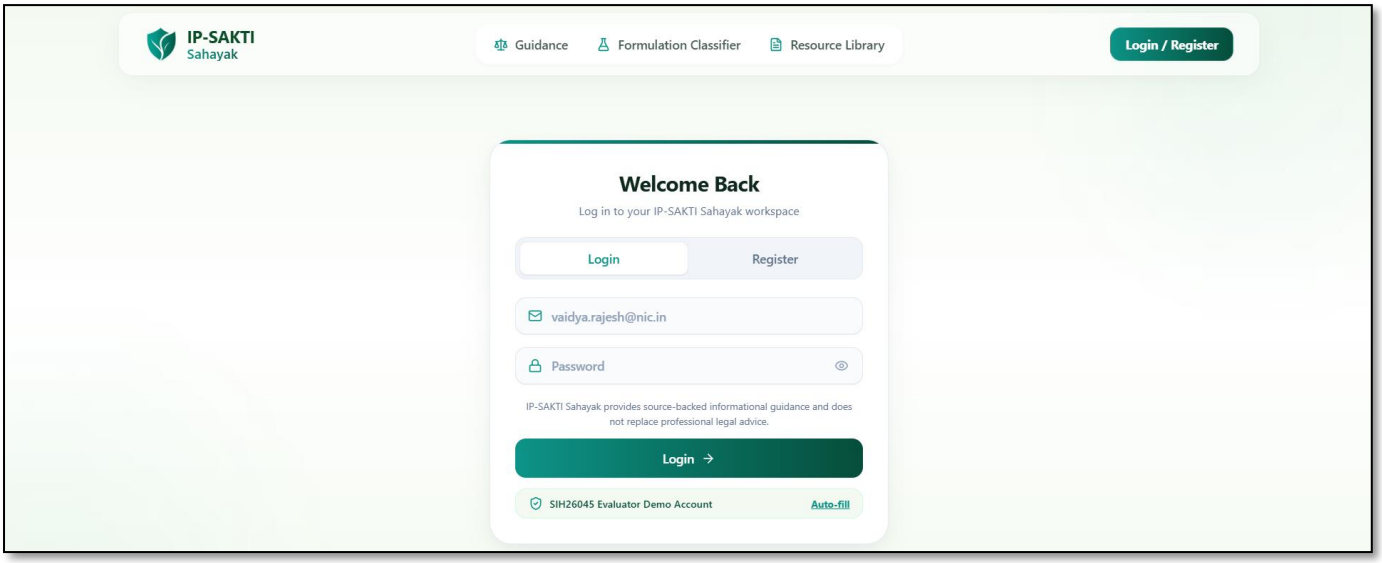


Figure 3.2: User Sign-Up and Login Interface



Figure 3.3: Regulatory Guidance Interface

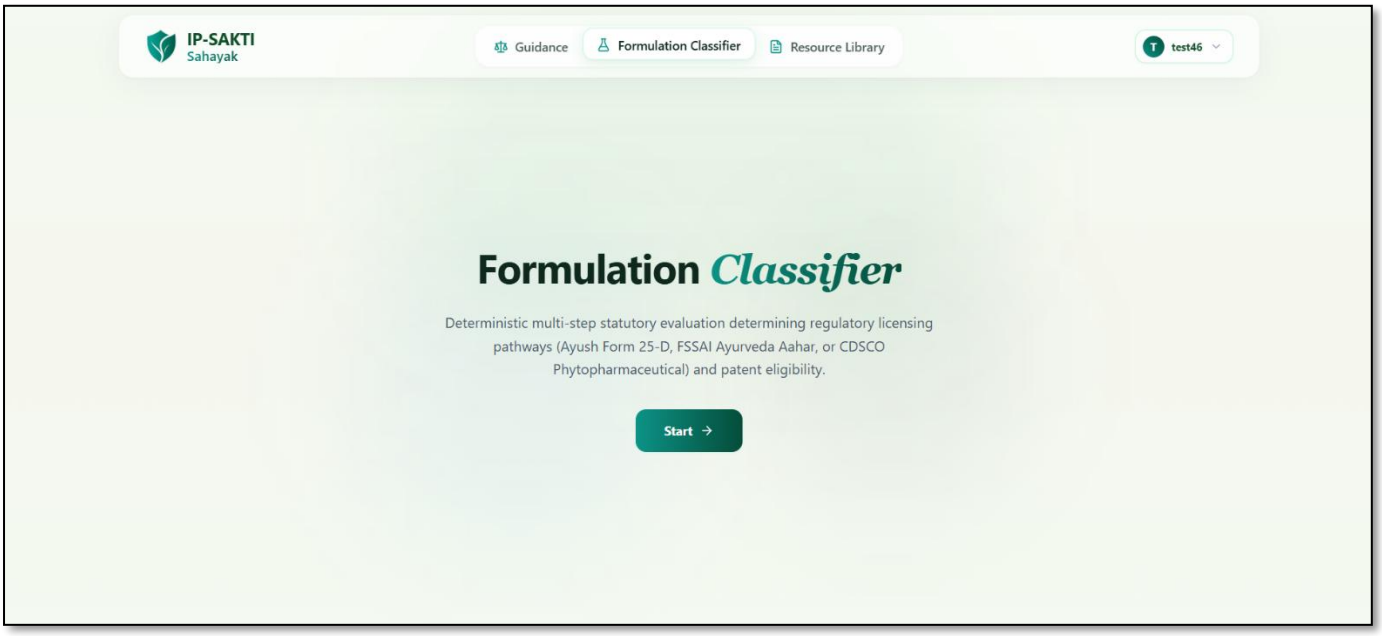


Figure 3.4: Formulation Classifier Interface

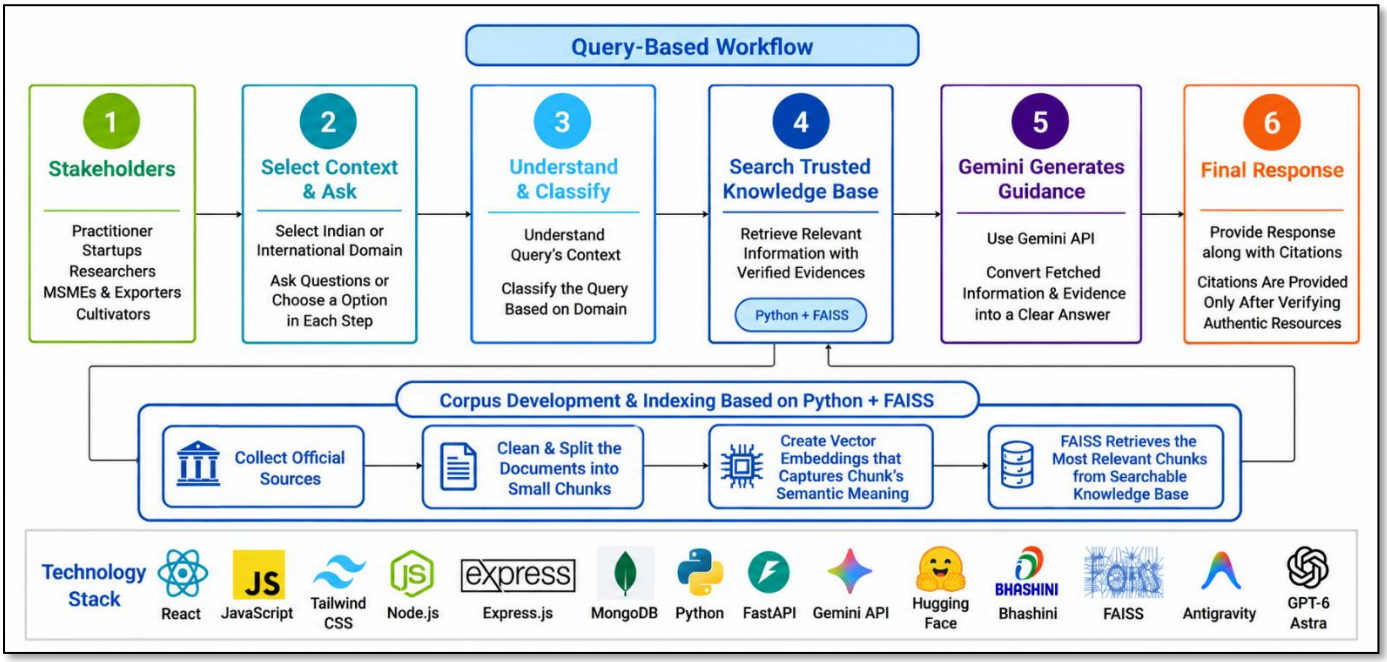


Figure 3.5: Query-Based System Architecture and Technology Stack

4. Complete Code

This section highlights key backend and frontend code for API setup, server initialization, and page navigation.

4.1 index.js

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const cookieParser = require("cookie-parser");
const corsOptions = require("./config/cors");
const errorHandler = require("./middlewares/error.middleware");

// Import Master V1 Router
const v1Router = require("./routes/index");

// Legacy route aliases for backward compatibility
const authRoutes = require("./routes/auth.routes");
const guidanceRoutes = require("./routes/guidance.routes");
const classifierRoutes = require("./routes/classifier.routes");
const resourceRoutes = require("./routes/resource.routes");
const historyRoutes = require("./routes/history.routes");
const dashboardRoutes = require("./routes/dashboard.routes");

const app = express();

// Database state mapping for in-memory connection inspection (O(1), zero DB I/O)
const DB_READY_STATES = {
  0: "disconnected",
  1: "connected",
  2: "connecting",
  3: "disconnecting",
};

// 1. Production-Grade Public Health Check (Render HTTP health checks, external monitors)
// Mounted before body parsers and auth to ensure maximum throughput and minimum latency
app.get("/health", (req, res) => {
  res.set("Cache-Control", "no-store, no-cache, must-revalidate, proxy-revalidate");
  res.set("Access-Control-Allow-Origin", "*");

  const dbState = DB_READY_STATES[mongoose.connection.readyState] || "disconnected";

  res.status(200).json({
    status: "ok",
    service: "IP-SAKTI Sahayak API",
    uptime: process.uptime(),
    timestamp: new Date().toISOString(),
    database: dbState,
  });
});

// 2. Global Middlewares
app.use(cors(corsOptions));
app.use(cookieParser());
```

```
app.use(express.json({ limit: "5mb" }));
app.use(express.urlencoded({ extended: true, limit: "5mb" }));

app.get("/api/health", (req, res) => {
  res.status(200).json({
    success: true,
    message: "IP-SAKTI Sahayak Backend API is running healthy.",
    timestamp: new Date().toISOString(),
  });
});

// 3. Mount Primary Versioned API Routes (/api/v1/*)
app.use("/api/v1", v1Router);

// 4. Backwards-Compatible Unversioned Routes (/api/*)
app.use("/api/auth", authRoutes);
app.use("/api/guidance", guidanceRoutes);
app.use("/api/classifier", classifierRoutes);
app.use("/api/resources", resourceRoutes);
app.use("/api/history", historyRoutes);
app.use("/api/dashboard", dashboardRoutes);

// 5. Global Centralized Error Handler
app.use(errorHandler);

module.exports = app;
```

Explanation: Sets up Express middleware and health checks, registers API routes, and applies centralized error handling.

4.2 server.js

```
require("dotenv").config();
const validateEnv = require("./config/env");
const connectDatabase = require("./config/db");
const app = require("./index");

async function startServer() {
  try {
    // 1. Validate environment configuration on boot
    const env = validateEnv();

    // 2. Connect to MongoDB database
    await connectDatabase();

    // 3. Warm up Python RAG Bridge in background (non-blocking)
    const { ensureRAGBridge, cleanupRAGBridge } = require("./services/rag.service");
    ensureRAGBridge().catch((err) => {
      console.warn("[Server] Initial RAG Bridge warmup check:", err.message);
    });

    // 4. Start HTTP server listener (explicitly bind to 0.0.0.0 for Render/Docker compatibility)
    const server = app.listen(env.PORT, "0.0.0.0", () => {
      console.log(
        `IP-SAKTI Sahayak Server is running on port ${env.PORT} [Environment: ${env.NODE_ENV}]`
```

```
      );
      console.log(`Allowed Frontend Origin: ${env.FRONTEND_URL}`);
    });

    // Graceful shutdown handling
    const handleShutdown = async (signal) => {
      console.log(`\nReceived ${signal}. Shutting down gracefully...`);
      cleanupRAGBridge();
      server.close(async () => {
        const mongoose = require("mongoose");
        await mongoose.disconnect();
        console.log("HTTP server and database connections closed.");
        process.exit(0);
      });
    };

    process.on("SIGTERM", () => handleShutdown("SIGTERM"));
    process.on("SIGINT", () => handleShutdown("SIGINT"));
  } catch (error) {
    console.error("Failed to start server:", error.message || error);
    process.exit(1);
  }
}

startServer();
```

Explanation: Validates environment settings, connects to MongoDB, warms up the RAG bridge, starts the server, and closes resources gracefully during shutdown.

4.3 App.jsx

```
import React, { useEffect, lazy, Suspense } from 'react';
import { BrowserRouter, Routes, Route, useLocation, Navigate, Outlet } from 'react-router-dom';
import { AuthProvider } from './context/AuthContext.jsx';
import { ToastProvider, ToastViewport } from './context/ToastContext.jsx';
import Navbar from './components/Navbar.jsx';
import Footer from './components/Footer.jsx';
import ProtectedRoute from './components/ProtectedRoute.jsx';
import Loader from './components/Loader.jsx';
import ScrollToTop from './components/ScrollToTop.jsx';

// Route-Level Code Splitting for optimal initial bundle & fast FCP/LCP
const LandingPage = lazy(() => import('./pages/LandingPage.jsx'));
const Login = lazy(() => import('./pages/Login.jsx'));
const Register = lazy(() => import('./pages/Register.jsx'));
const Dashboard = lazy(() => import('./pages/Dashboard.jsx'));
const Guidance = lazy(() => import('./pages/Guidance.jsx'));
const FormulationClassifier = lazy(() => import('./pages/FormulationClassifier.jsx'));
const ResourceLibrary = lazy(() => import('./pages/ResourceLibrary.jsx'));
const History = lazy(() => import('./pages/History.jsx'));
const HowItWorksPage = lazy(() => import('./pages/HowItWorksPage.jsx'));
const JurisdictionsPage = lazy(() => import('./pages/JurisdictionsPage.jsx'));
const IPRCoveragePage = lazy(() => import('./pages/IPRCoveragePage.jsx'));
const NotFound = lazy(() => import('./pages/NotFound.jsx'));
```

```
function ScrollHandler() {
  const { pathname, hash } = useLocation();

  useEffect(() => {
    if (hash) {
      const targetId = hash.replace('#', '');
      let attempts = 0;

      const performScroll = (smooth = true) => {
        const el = document.getElementById(targetId);
        if (el) {
          const nav = document.querySelector('header');
          const navHeight = nav ? nav.getBoundingClientRect().height : 72;
          const rect = el.getBoundingClientRect();
          const targetY = window.pageYOffset + rect.top - navHeight;

          window.scrollTo({
            top: Math.max(0, targetY),
            behavior: smooth ? 'smooth' : 'instant'
          });
          return true;
        }
        return false;
      };

      // Try immediate scroll
      performScroll(false);

      // Keep observing until element is found and layout stabilizes
      const timer = setInterval(() => {
        attempts++;
        const el = document.getElementById(targetId);
        if (el) {
          const nav = document.querySelector('header');
          const navHeight = nav ? nav.getBoundingClientRect().height : 72;
          const currentTop = Math.round(el.getBoundingClientRect().top);
          // If not aligned right under the navbar (due to lazy route/image loading), adjust
          if (Math.abs(currentTop - navHeight) > 4) {
            const targetY = window.pageYOffset + el.getBoundingClientRect().top - navHeight;
            window.scrollTo({
              top: Math.max(0, targetY),
              behavior: attempts <= 3 ? 'instant' : 'smooth'
            });
          } else {
            clearInterval(timer);
          }
        }
        if (attempts > 30) clearInterval(timer);
      }, 60);

      return () => clearInterval(timer);
    } else {
      window.scrollTo({ top: 0, left: 0, behavior: 'instant' });
    }
  }, [pathname, hash]);
```

```
  return null;
}

function PageFallback() {
  return (
    <div className="min-h-[60vh] flex items-center justify-center">
      <Loader message="Loading workspace..." detail="Preparing statutory intelligence..." />
    </div>
  );
}

function AppLayout() {
  return (
    <div   className="flex   flex-col   min-h-screen   bg-[#F4FAF1]   text-[#0D1B2A]   font-sans
selection:bg-[#1E7466]/20 selection:text-[#145348]">
      <Navbar />
      <main className="flex-1 w-full">
        <Suspense fallback={<PageFallback />}>
          <Outlet />
        </Suspense>
      </main>
      <Footer />
      <ScrollToTop />
    </div>
  );
}

export default function App() {
  return (
    <ToastProvider>
      <AuthProvider>
        <BrowserRouter>
          <ScrollHandler />
          <Routes>
            {/* Standard Application Shell (Navbar, Main, Footer, ScrollToTop) */}
            <Route element={<AppLayout />}>
              {/* Public Discovery Landing & Informational Pages */}
              <Route path="/" element={<LandingPage />} />
              <Route path="/about" element={<Navigate to="/#about" replace />} />
              <Route path="/faq" element={<Navigate to="/#faq" replace />} />
              <Route path="/how-it-works" element={<HowItWorksPage />} />
              <Route path="/jurisdictions" element={<JurisdictionsPage />} />
              <Route path="/ipr-coverage" element={<IPRCoveragePage />} />

              {/* Authentication */}
              <Route path="/login" element={<Login />} />
              <Route path="/register" element={<Register />} />

              {/* Primary Workspaces — Guidance & Classifier are Protected; Resource Library
remains Public */}
              <Route
                path="/guidance"
                element={
                  <ProtectedRoute>
                    <Guidance />
```

```
                </ProtectedRoute>
              }
            />
            <Route
              path="/classifier"
              element={
                <ProtectedRoute>
                  <FormulationClassifier />
                </ProtectedRoute>
              }
            />
            <Route path="/resources" element={<ResourceLibrary />} />

            {/* Protected Workspace Views */}
            <Route
              path="/dashboard"
              element={
                <ProtectedRoute>
                  <Dashboard />
                </ProtectedRoute>
              }
            />
            <Route
              path="/history"
              element={
                <ProtectedRoute>
                  <History />
                </ProtectedRoute>
              }
            />
          </Route>

          {/* Standalone Edge-to-Edge 404 Experience — No Navbar, No Footer */}
          <Route
            path="*"
            element={
              <Suspense fallback={<PageFallback />}>
                <NotFound />
              </Suspense>
            }
          />
        </Routes>
      </BrowserRouter>
    </AuthProvider>
  </ToastProvider>
 );
}
```

Explanation: Configures React routes and shared layout, lazy-loads pages, protects selected routes, and handles scrolling and not-found pages.

5. Output Screenshots



Figure 5.1 — Regulatory Guidance Query Interface

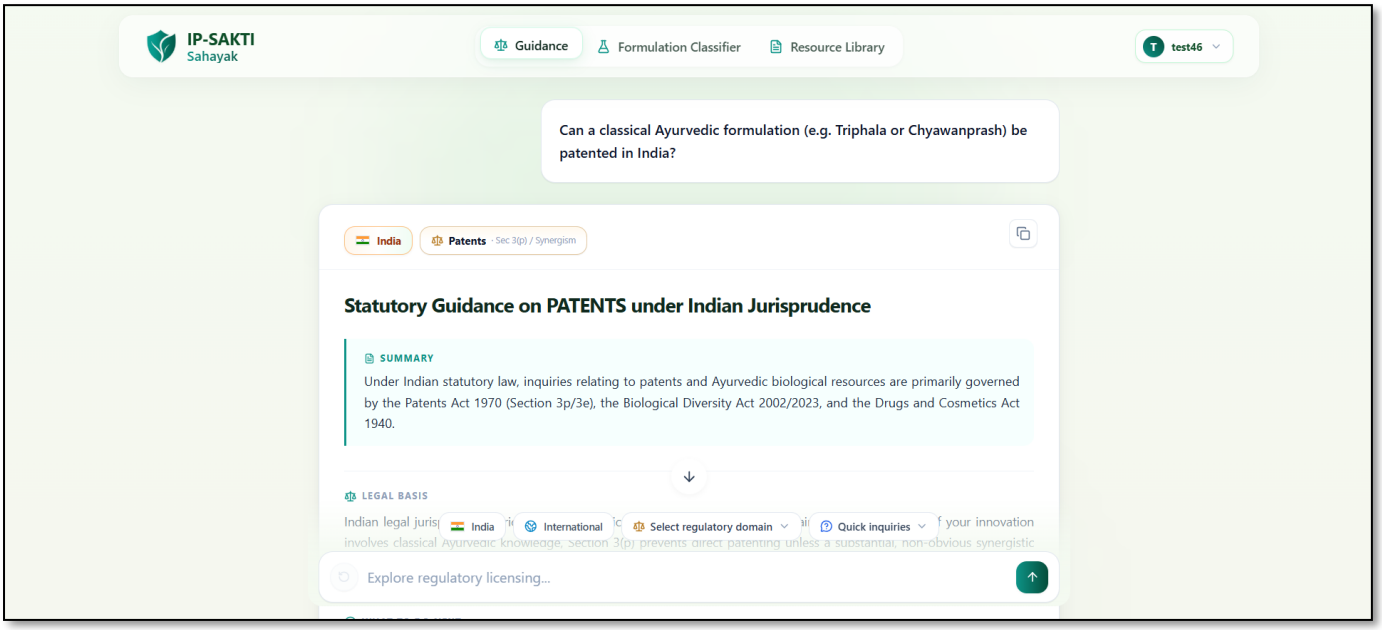


Figure 5.2 — Structured Guidance Response

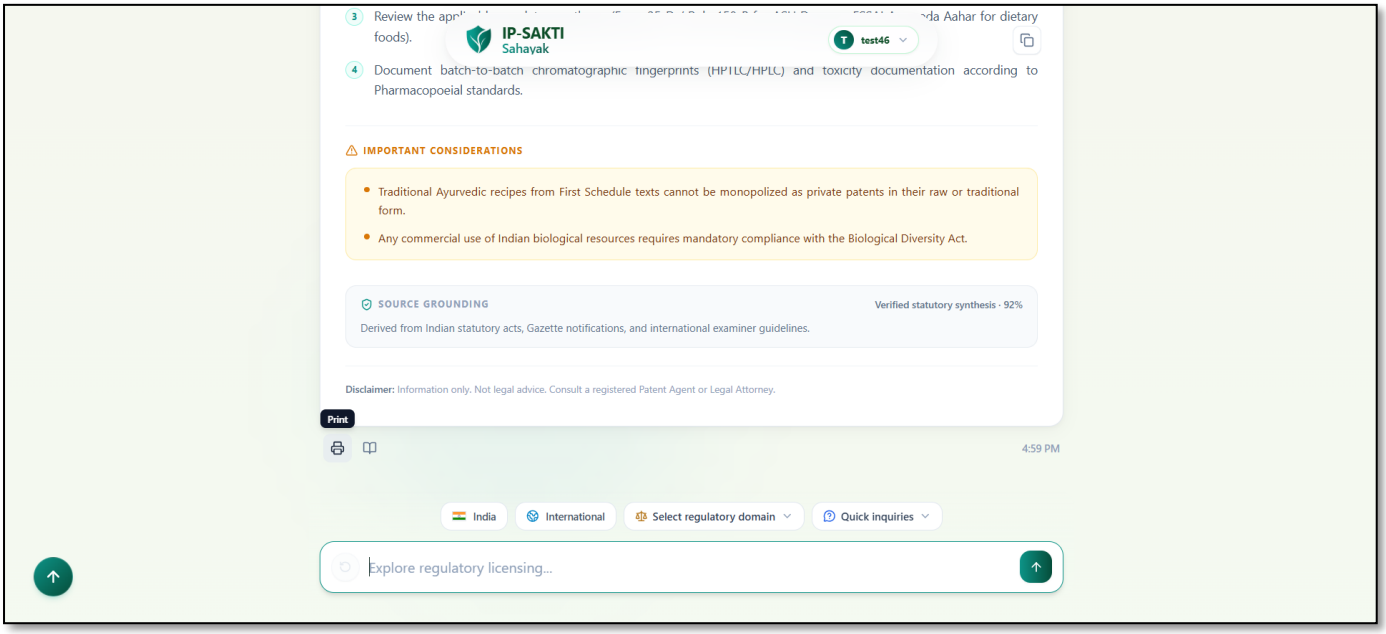
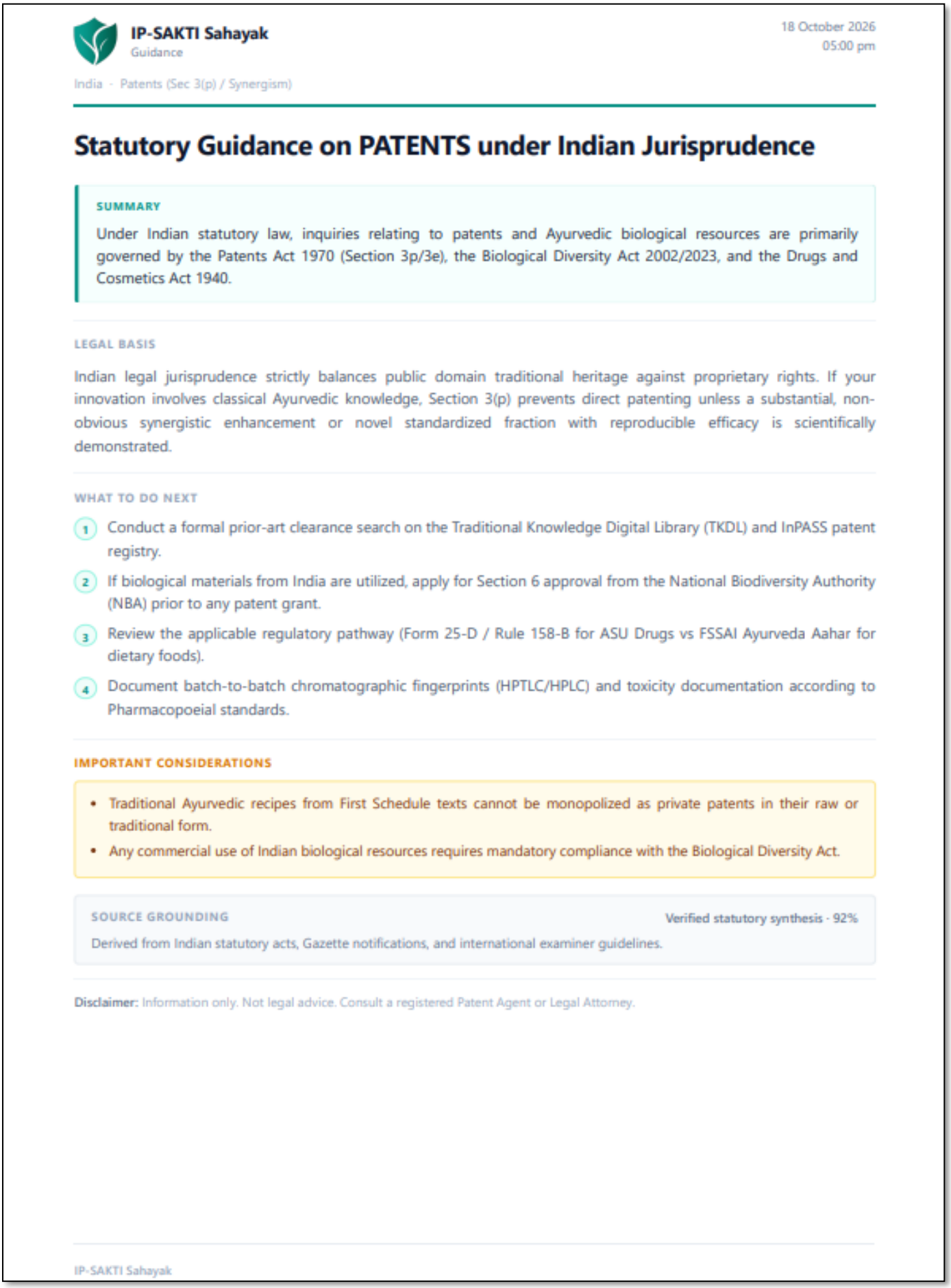


Figure 5.3 — Guidance Details and Source Grounding



IP-SAKTI Sahayak
Guidance

18 October 2026
05:00 pm

India · Patents (Sec 3(p) / Synergism)

Statutory Guidance on PATENTS under Indian Jurisprudence

SUMMARY

Under Indian statutory law, inquiries relating to patents and Ayurvedic biological resources are primarily governed by the Patents Act 1970 (Section 3p/3e), the Biological Diversity Act 2002/2023, and the Drugs and Cosmetics Act 1940.

LEGAL BASIS

Indian legal jurisprudence strictly balances public domain traditional heritage against proprietary rights. If your innovation involves classical Ayurvedic knowledge, Section 3(p) prevents direct patenting unless a substantial, non-obvious synergistic enhancement or novel standardized fraction with reproducible efficacy is scientifically demonstrated.

WHAT TO DO NEXT

1. Conduct a formal prior-art clearance search on the Traditional Knowledge Digital Library (TKDL) and InPASS patent registry.
2. If biological materials from India are utilized, apply for Section 6 approval from the National Biodiversity Authority (NBA) prior to any patent grant.
3. Review the applicable regulatory pathway (Form 25-D / Rule 158-B for ASU Drugs vs FSSAI Ayurveda Aahar for dietary foods).
4. Document batch-to-batch chromatographic fingerprints (HPTLC/HPLC) and toxicity documentation according to Pharmacopoeial standards.

IMPORTANT CONSIDERATIONS

- Traditional Ayurvedic recipes from First Schedule texts cannot be monopolized as private patents in their raw or traditional form.
- Any commercial use of Indian biological resources requires mandatory compliance with the Biological Diversity Act.

SOURCE GROUNDING — Verified statutory synthesis - 92%

Derived from Indian statutory acts, Gazette notifications, and international examiner guidelines.

Disclaimer: Information only. Not legal advice. Consult a registered Patent Agent or Legal Attorney.

IP-SAKTI Sahayak

Figure 5.4 — Printable Guidance Report

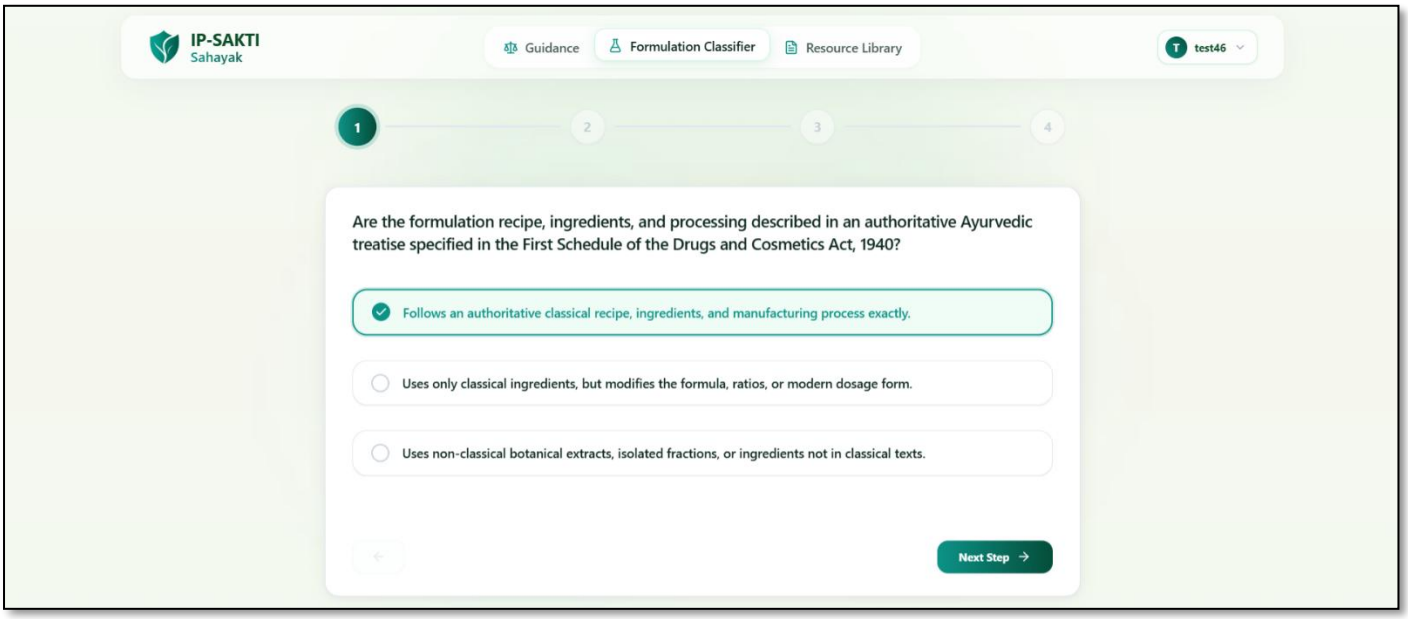


Figure 5.5: Textual Origin Selection

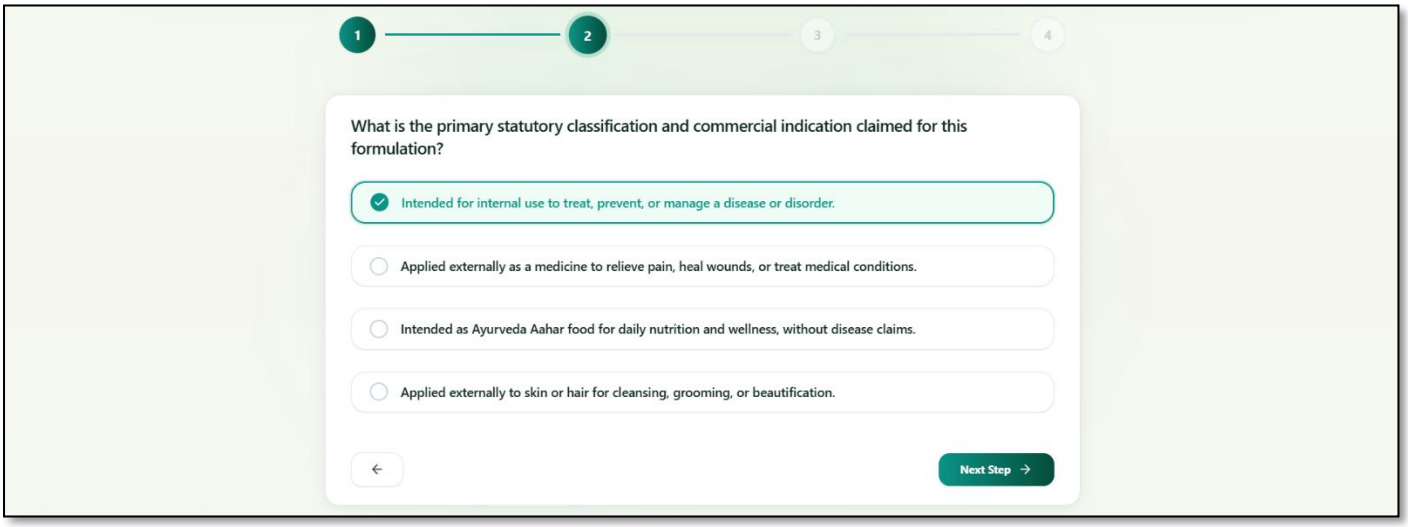


Figure 5.6: Intended Use Selection

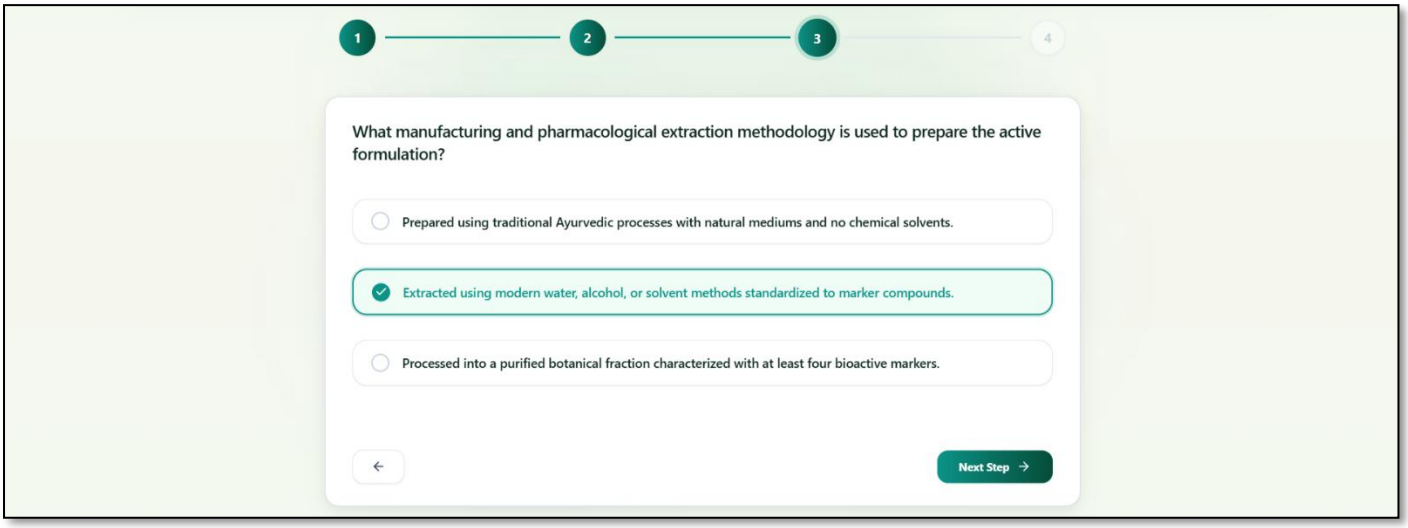


Figure 5.7: Processing and Extraction Method

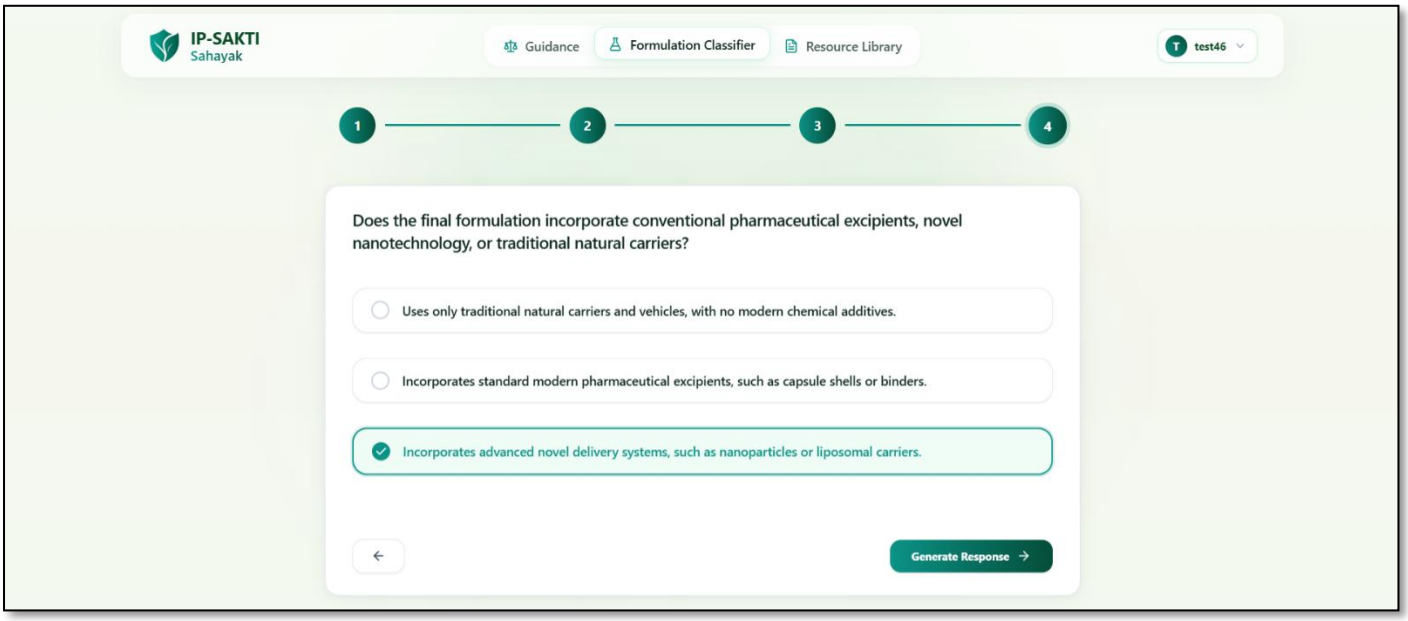


Figure 5.8: Excipients and Delivery System

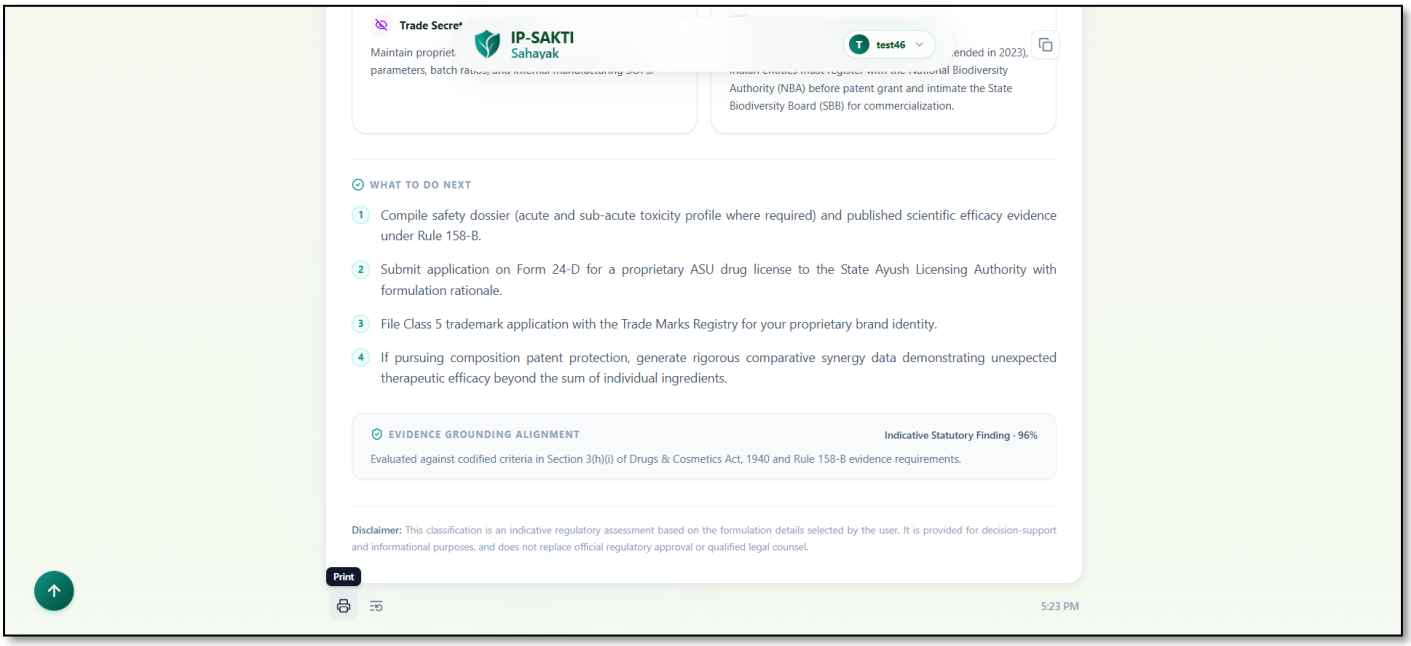


Figure 5.9: Classification Results Dossier



IP-SAKTI Sahayak

18 October 2026
05:26 pm

Classification Dossier

Patent or Proprietary (P&P) Ayurvedic Medicine

SUMMARY

Your formulation is categorized as a Patent or Proprietary Ayurvedic medicine containing ingredients from First Schedule treatises in a proprietary ratio, combination, or modified delivery form.

LEGAL BASIS

All active ingredients are recorded in First Schedule authoritative treatises, but the formula differs from classical recipes (altered proportions, modern extracts, or modern dosage forms) and is not administered parenterally.

REGULATORY PATHWAY

MANUFACTURING LICENSE FORM
Form 25-D (Application on Form 24-D / Loan License on Form 25-E via Form 24-E)

LICENSING AUTHORITY
State Ayush Licensing Authority (SLA)

CLINICAL EVIDENCE MANDATE
Requires safety and efficacy evidence under Rule 158-B: published literature and pilot clinical trial / acute toxicity data depending on whether indications or extract ratios are novel.

GOOD MANUFACTURING PRACTICES (GMP)
Mandatory Schedule T GMP compliance

LABELING STANDARD
Must display distinctive proprietary brand name, full ingredient list with Latin binomials, true formula disclosure, and dosage as per Rule 161.

INTELLECTUAL PROPERTY OPPORTUNITIES & CONSTRAINTS

PATENT FEASIBILITY (SECTION 3P/3E)
Patentable in India only if unexpected synergistic technical effect is demonstrated by comparative bio-assays to overcome Section 3(e) (mere admixture) and Section 3(p) (traditional knowledge).

TRADEMARK STRATEGY
Strong candidate for Class 5 brand name and logo trademark registration.

TRADE SECRETS & KNOW-HOW
Maintain proprietary protection over standardized extraction parameters, batch ratios, and internal manufacturing SOPs.

NBA / ABS COMPLIANCE OBLIGATION
Under the Biological Diversity Act, 2002 (as amended in 2023), Indian entities must register with the National Biodiversity Authority (NBA) before patent grant and intimate the State Biodiversity Board (SBB) for commercialization.

WHAT TO DO NEXT

1. Compile safety dossier (acute and sub-acute toxicity profile where required) and published scientific efficacy evidence under Rule 158-B.
2. Submit application on Form 24-D for a proprietary ASU drug license to the State Ayush Licensing Authority with formulation rationale.
3. File Class 5 trademark application with the Trade Marks Registry for your proprietary brand identity.
4. If pursuing composition patent protection, generate rigorous comparative synergy data demonstrating unexpected therapeutic efficacy beyond the sum of individual ingredients.

EVIDENCE GROUNDING ALIGNMENT — Indicative Statutory Finding - 96%
Evaluated against codified criteria in Section 3(h)(i) of Drugs & Cosmetics Act, 1940 and Rule 158-B evidence requirements.

Disclaimer: This classification is an indicative regulatory assessment based on the formulation details selected by the user. It is provided for decision-support and informational purposes, and does not replace official regulatory approval or qualified legal counsel.

IP-SAKTI Sahayak — Formulation Classification Dossier

Figure 5.10: Printable Assessment Report

6. Observations

1. The interface allows users to select India or International and choose a regulatory domain before entering their question using the text input field and submission button.

2. The guidance response presents information in structured sections such as Summary and Legal Basis to help users understand the main findings and their supporting legal reasoning.

3. The response includes a What To Do Next section with numbered suggestions and an Important Considerations section highlighting relevant regulatory points.

4. Source-grounding information and a legal disclaimer are displayed to communicate the stated basis and limitations of the guidance. The print control allows users to prepare a printable report for reference.

5. The chat-style interface supports successive regulatory inquiries within the same conversation. The displayed response is demonstration content and requires verification after integration with the live RAG service.

6. The first step asks whether the formulation follows an authoritative Ayurvedic treatise specified in the First Schedule of the Drugs and Cosmetics Act, 1940. Users can select an exact classical formulation or indicate modifications or non-classical ingredients before proceeding.

7. The second step collects information about the formulation's intended use. Users can select internal therapeutic use and external application alongside Ayurveda Aahara for dietary purposes or topical cosmetic use.

8. The third step asks how the formulation is manufactured or extracted. Users can select traditional Ayurvedic preparation methods or indicate modern extraction techniques or purified botanical fractions.

7. Conclusion

The IP-SAKTI Sahayak mini-project successfully develops a source-grounded Retrieval-Augmented Generation (RAG) pipeline for intellectual property and regulatory guidance in Ayurveda. Using a curated corpus of 11 legal documents and 100 physical PDF pages, the system combines data preprocessing and MiniLM embeddings with FAISS retrieval and Gemini-based response generation to provide structured guidance supported by traceable legal evidence.

The system achieved 166 passing offline tests and demonstrated successful source retrieval and citation validation on the evaluated cases. Although the core RAG pipeline has been developed and tested, complete frontend-backend integration and broader evaluation remain pending. Overall, the project demonstrates how data science and generative AI can support accessible and evidence-based regulatory guidance while acknowledging the limitations of the available corpus.